



Evaluating PlanTEA: the practice of a UX evaluation methodology for people with ASD

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Abstract

Providing user-friendly software for groups with special needs is crucial. However, there is a lack of specific evaluation methods and techniques adapted to the characteristics and needs of some users, such as those with autism spectrum disorder (ASD). The main objective of this work is to validate a UX evaluation methodology that focuses specifically on the use of systems, products, and services by adults with ASD through a specific use case. This methodology was developed by taking into account the perspectives of UX experts and ASD professionals on the one hand and potential users on the other. It comprises a three-stage validated process based on the UX/ASD model. It focuses on assessing the user experience of systems used by adults with ASD by considering specific tools and techniques for effective evaluation, in contrast to the existing UX methodologies for evaluating software applications for users with ASD that are limited to the use of generic usability analysis tools. The choice of the PlanTEA application is motivated by the fact that it is designed specifically for users with ASD to assist them in their visits to medical centers by planning and anticipating those situations. However, it has not been evaluated from the ASD user experience point of view. Applying this specific methodology for users with ASD has provided significant feedback for improving their satisfaction and perception of PlanTEA before making it available to society. Moreover, the methodology has been proven to be a powerful tool in the whole process of the UX evaluation of PlanTEA, allowing for more comprehensive UX assessments and reporting.

Keywords Autism spectrum disorder · Assistive technologies · User experience evaluation · Methodology

1 Introduction

Autism spectrum disorder (ASD) is a neurological dysfunction of genetic origin that manifests early in infancy and lasts throughout time. Its symptoms impair social interaction, communication, and flexibility of behavior and thought

[1]. It is among the most prevalent neurological conditions in young people, and the number of diagnoses has considerably increased in recent years. Although prevalence rates vary by nation, the World Health Organization estimates that one in every 100 children worldwide has ASD.

Since ASD exhibits a variety of symptoms, it is considered a spectrum. No two people with ASD are alike due to the heterogeneity of this disorder, and symptoms can change depending on the severity. This variability can be affected by several factors, including the level of support the individual

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receives, language development, and additional associated disorders or illnesses, such as intellectual disability or psychiatric disorders [2]. As a result, it is critical to treat each patient individually and offer them specialized support that considers their needs and circumstances [3].

Numerous technology developments and applications, particularly those that support touch and mobile devices, have been developed for people with ASD, and have demonstrated their efficacy in interventions and skill development [4]. In addition to offering a motivating and highly predictable environment, these assistive technologies can provide individualized support tailored to each individual's communication needs, sensory challenges, preferences, and interests [5].

There is a growing interest in the use of new technologies in the intervention and support of users with ASD, with a considerable increase in developments oriented to this group of users. However, most of these systems have not been appropriately evaluated from the point of view of user experience [6]. User experience (UX) is defined by the standard ISO 9241-210 as “*user's perceptions and responses that result from the use and/or anticipated use of a system, product or service*” [7]. In other words, the user experience is the degree of “satisfaction” that the end user has with the system or service after using it, that is based on each of the interactions that he or she has. A part of UX is usability, defined by the ISO 9241-11 [8] standard as “*the extent to which a system, product or service can be used by specific users to achieve specific goals with effectiveness, efficiency and satisfaction in a specific context of use*”. The concept of usability is related to the fulfillment of tasks and the satisfaction experienced by users. Therefore, a higher degree of usability of a system, product, or service after user interaction leads to a better UX.

Evaluating the usability and, more generally, the UX of applications oriented to users with ASD is challenging. Only some existing developments evaluate UX, and the few that have done so have been limited to generic usability analysis tools, such as adaptations of the Nielsen heuristics [9] or informal interviews with family members and caregivers. Therefore, there is a need to properly assess and evaluate this type of development with specific methods and instruments [10].

To address the need for anticipation and communication identified in the autistic population, the PlanTEA app was created [11]. This application, designed for tablet devices, allows ASD users and their caregivers or therapists to plan and anticipate visits to medical centers based on an augmentative and alternative communication system. The application has been developed to address the limitations of existing ones, thus providing a more significant set of functionalities than any other, as well as including a specific communication notebook for individuals with ASD, which facilitates their interaction with healthcare personnel.

In contrast to most existing developments in the literature, PlanTEA's design process followed specific design principles for ASD users (AutismGuide) [12], facilitating the creation and implementation of a more suitable software solution for ASD individuals.

The application was subjected to a preliminary evaluation in which 17 subjects with different profiles (family members and caregivers of people with ASD, specialists, therapists, and software developers with expertise in usability) participated. The results of this evaluation were very positive [11]. Still, it had two limitations: the first is that only generic usability evaluation instruments were used, such as the SUS (System Usability Scale) [13] or the TAM (Technology Acceptance Model) framework [14]; the second, is that people with ASD, who constitute the main end users, did not participate. Therefore, a new evaluation of PlanTEA using a methodology specifically designed for assessing the user experience of individuals with ASD and employing concrete instruments tailored to this population could provide valuable insights. On one hand, it represents an excellent opportunity to apply and validate the methodology developed by Valencia et al. [15]. On the other hand, the results will provide valuable feedback for the improvement of PlanTEA. Furthermore, conducting more than one evaluation is always beneficial because there is no perfect evaluation method.

The rest of the paper is organized as follows: Sect. 2 summarizes the previous work related to the design and evaluation of software applications for ASD users, focusing on the recent methodology proposal to evaluate the user experience in applications oriented to ASD people; then, Sect. 3 describes the PLANTEA application, and Sect. 4 explains the process followed to assess the user experience. Finally, the conclusions are presented in Sect. 5.

2 Design and evaluation of applications for ASD users

Currently, technology support for people with ASD is of increasing interest [6]. New technologies are well accepted by this collective, who feel motivated and comfortable using this environment, as it is highly predictable for them. Existing applications focus on assisting mainly in the areas of socialization [16, 17], communication [18], and emotion recognition [19, 20]. The objective of all of them is to enhance and train the skills of people with ASD to promote their inclusion and quality of life, as well as that of their families [21].

Most existing developments are in the mobile computing paradigm [22, 23] and usually present a serious game format [18, 24]. More recently, the use of advanced interaction styles and paradigms, such as Virtual and Augmented Reality [25, 26], assistive robots [27], or wearable devices [28], has begun to be exploited. However, few systems have been

evaluated, and those tested did not use any methodology or instrument specifically oriented to assess the user experience of ASD users [6].

Applications for people with ASD, such as those discussed in the previous paragraph, are usually created ad-hoc, with no guidelines or frameworks to assist designers and developers during the creation and evaluation processes. Recently, new approaches have been proposed for developing and assessing applications specifically oriented to ASD. These approaches suggest more guided and structured development processes, and some incorporate guidelines, recommendations, evaluation methods, and tools. Most design methodologies focus on creating serious games [29, 30] and propose a set of phases to follow in making this specific type of application. Regarding particular design guidelines for applications aimed at people with ASD, the Autism Guide usability guidelines [12] offers a comprehensive approach to technology accessibility for individuals with autism, encompassing various recommendations and considerations. These guidelines stand out since they are based on a literature review focusing on the unique needs that ASD people may have and that differ from those with cognitive or learning disabilities, including distinct symptoms, comorbidities, and sensory and motor characteristics. A total of 69 recommendations are proposed, which are classified into 11 categories: (1) general usability principles, (2) nonfunctional requirements, (3) functional requirements, (4) adaptability, (5) guidance, (6) workload, (7) compatibility, (8) explicit control, (9) significance of codes, (10) error management, and (11) consistency.

There are also design recommendations for specific paradigms, such as Virtual Reality [31] or mobile computing [32].

Regarding the existing methodologies to evaluate the applications developed for ASD users, a recent review of the related work [15] has identified that most studies have employed modified versions of the SUS instrument [13] to gauge usability, adapting it for ASD users by simplifying language, incorporating emoticons, or reducing the scale size. Additionally, evaluations have utilized heuristic evaluations, questionnaires, and a combination of methods, including direct observation, to comprehensively assess usability and user experience. Other evaluation methods for ASD users are based on verbal communication or hand simulation, which while useful for gathering direct user feedback, may present challenges due to potential communication deficits and frustration among ASD users, necessitating tailored approaches for their implementation. The review has concluded that empirical assessment experiences should provide more detailed specifications regarding the evaluations conducted, ensuring that the methods and tools employed are tailored to accommodate the unique characteristics and requirements of individuals with ASD. Furthermore, the

active involvement of a diverse group including UX/Usability experts, ASD specialists, educators, and individuals with ASD themselves is essential.

Consequently, the proposal described in [15] appears to be the most comprehensive, as it considers these issues to define a specific methodology for evaluating the experience of ASD users. This methodology addresses the need for a formal process to evaluate the UX of systems, products, or services used by people with ASD by considering the necessary tools and techniques for effective evaluation. It utilizes a validated process with specific methods and instruments selected and adapted according to the needs of people with ASD. Employing the suggested approach and a suitable choice or modification of tools, as recommended in this study, can enhance the satisfaction and perception of individuals with ASD regarding the evaluated system. This proposal is based on specific UX factors in interactive applications for ASD users [10], including a checklist for evaluating compliance with the UX design principles [33], recommendations and suggestions for user testing, a set of questions for the Perspective-Based Inspection method, and a set of heuristics. The following subsection provides more details on the main features of this methodological approach.

2.1 A methodology to evaluate the user experience in applications oriented to ASD users

Valencia et al. [15] have proposed a methodology to evaluate the user experience in applications tailored to ASD users. This methodology comprises a three-stage validated process based on the UX/ASD model, characterized by nine distinct UX factors: engaging, predictable, structured, interactive, generalizable, customizable, sense-aware, attention retaining, and frustration free [10, 13]. The underpinnings of the UX evaluation methodology rely on this model in conjunction with the characteristics of ASD users.

Such methodology focuses on assessing the user experience of systems, products, or services utilized by children and adults with ASD in accordance with the DSM5 [34]. It tries to gather as much helpful information as possible regarding the user experience (UX) of the system, product, or service, to use that information to improve the UX and make the system, product, or service more usable by people with ASD. The three stages, which run consecutively, are the Planning Stage (S1), the Execution Stage (S2), and the Results Analysis Stage (S3). Figure 1 gives a general overview of the process.

The *Planning* stage (S1) aims to define the UX assessments to be performed and to find UX/Usability experts, domain experts (professionals working with people with ASD), participants, and tutors. It comprises the selection and planning of the evaluation methods to be executed (S1.1), the definition of the scenarios, tasks, and protocols

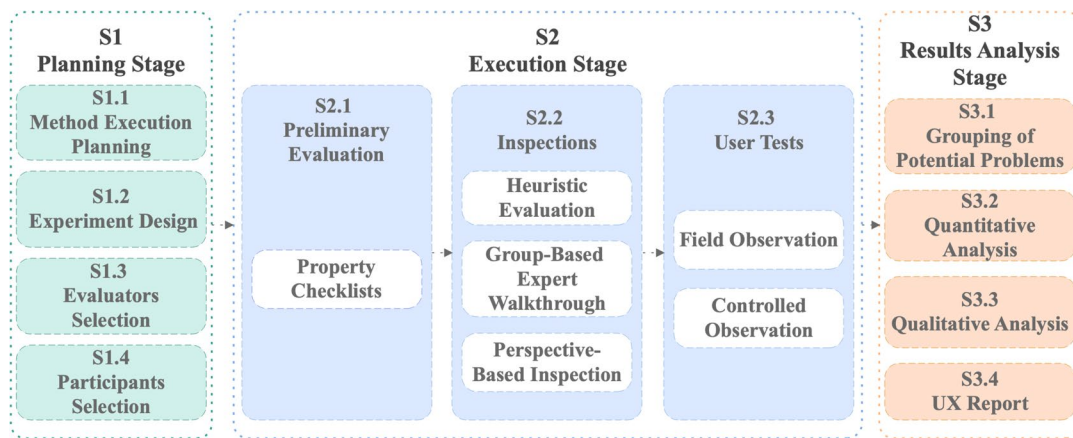


Fig. 1 Stages of the UX evaluation methodology for people with ASD

to be used during the evaluation (S1.2), the selection of UX experts and specialists in the specific domain to participate in the evaluation (S1.3), and the selection of at least three to five users who participate in the evaluation (S1.4).

The purpose of the *Execution* stage (S2) is to complete the methods previously selected in the planning stage. To this end, a preliminary evaluation (S2.1) is proposed to detect deficiencies in the system through a property checklist based on nine UX factors for people with ASD previously enumerated. This method utilizes a checklist to evaluate design properties, ensuring alignment with intended objectives, encompassing both high-level factors like consistency and low-level details such as color and font size. Its purpose is to enable expert usability evaluators to verify the product's design conformity to checklist criteria.

Then, three inspection methods to identify potential usability problems and solutions are considered (S2.2), according to three different points of view: UX experts (group-based expert walkthrough), type of users (perspective-based inspection) and heuristics (heuristic evaluation). The group-based expert walkthrough relies on completing a series of tasks or scenarios led by a facilitator. It can use domain-specific criteria understood by professionals unfamiliar with UX. Ideally, someone proficient in both UX/Usability and the domain, or solely a UX/Usability expert, should lead the evaluation. The perspective-based inspection focuses on identifying usability issues through three user perspectives: novice, expert, and error-handling. Evaluators take on each user's role and viewpoint, guided by specific inspection questions tailored to each perspective, suggesting consideration of nine UX factors and perspectives including novice and expert users with ASD. Collaboration between ASD domain experts and UX/Usability experts is advised, as domain experts may better understand ASD user experiences. The heuristic evaluation entails inspectors scrutinizing a system for potential issues according to predetermined sets of heuristics, detailing severity, frequency,

and criticality. The incorporation of tasks and scenarios in this method is discretionary, subject to the discretion of the researchers. The execution stage ends with at least one user test (S2.3) to measure the satisfaction of the potential users with the system. It can be carried out by field observation. At the same time, the user interacts with the system by controlled observations, i.e., the execution of guided activities, to eliminate the noise of the data obtained by including strict controls.

Finally, the Results Analysis stage (S3) tries to organize the information obtained in the previous stage by grouping the main problems found (S3.1), performing both a quantitative and qualitative analysis (S3.2 and S3.3) of the obtained results, and producing a report (S3.4) containing the main problems encountered and the solutions proposed to improve the UX of the system. The collective evaluation of compliance across the suggested categories is preferred. To scrutinize the results, it is advised to compute and illustrate the category-wise compliance percentages, along with determining the overall compliance percentage post-assessment, thereby facilitating a comprehensive comprehension of the outcomes following the completion of the property checklist. Radar charts serve as a visually effective method for representing the acquired results.

The suggested methodology has undergone validation by experts in UX/Usability and specialists in ASD, who have deemed it highly beneficial and potentially supportive of the inclusion of individuals with ASD. Additionally, all involved participants have expressed their intention to employ it in the future.

3 PlanTEA

For individuals with ASD, facing unfamiliar environments like medical appointments can induce anxiety. Then, preparation is crucial, involving a detailed explanation of

procedures requiring higher-level cognitive functions. Anticipation boards aid in activity preparation and highlight cognitive flexibility as a critical skill. While technology offers benefits for learning and skill enhancement, existing systems often overlook the specific needs of ASD users. The review of existing literature reveals a focus on software solutions for ASD primarily geared towards cognitive training and emotion recognition [6]. These solutions commonly utilize mobile devices and incorporate gamification techniques for motivation [23, 24]. However, many need more free access to all features and support for multiple user roles. While mobile applications are recognized for promoting healthcare delivery and healthy behaviors, few address the specific needs of people with ASD in managing preparation and anticipation for routine medical appointments, with limited evaluation of their effectiveness. Addressing this gap, an intuitive tablet app to assist caregivers in planning medical visits, aiming to reduce stress for ASD individuals, has been developed, following a user-centered design approach and incorporating ASD-specific guidelines. The app, named PlanTEA [11], incorporates entertainment mechanisms during waiting times and positive reinforcement methods (rewards) and facilitates, thanks to the inclusion of an augmentative and alternative communication booklet, communication with medical staff when necessary.

A User-Centered Design (UCD) approach was employed during its development, focusing on visual metaphors, and minimizing distractions. This involved active participation from experts in mobile app usability and occupational therapy for ASD and input from families of affected children. Low-fidelity prototypes were iteratively refined with feedback from stakeholders. The graphical interface was designed with attention to usability and accessibility for parents/caregivers and children with ASD, accommodating verbal and non-verbal communication. AutismGuide usability guidelines [12] were considered in the design process and tailored to the application's objectives of planning and communication [11]. PlanTEA was developed for Android tablets using Android Studio and underwent formative evaluations for functional and mobile usability, including chromatic tests and careful control placement for users with motor impairments.

The PlanTEA application (Fig. 2) was created to assist the need for anticipation and communication for children and adults with ASD in a crucial activity. Initially, the domain supported is medical visits, as these are essential and inevitable and can cause a lot of anxiety for both the child and the family. However, future versions of the app will support other domains. The user interface is currently only available in Spanish, although other languages are planned to be supported in the future.

A planning and anticipation system and an augmentative and alternative communication system (AACs) are the

two major capabilities of PlanTEA (Fig. 2a), based on pictograms, and may be used by both verbal and non-verbal people with ASD. The house and a hospital are the two primary settings in this program. The application supports two different user types or roles: planner, for family members or caregivers, who oversee creating and monitoring the plans, and ASD person, who can read the chosen plans and access the communication notebook. Planners can organize and plan the steps of the doctor's appointment at home (Fig. 2b), identifying some rewards and activities the ASD person can engage in while waiting. The planning will be done with the help of pictograms related to the medical domain (places, persons, objects, and common actions in medical appointments). Still, it will also be possible to import external resources from the internet or use the device's camera (Fig. 2d). At home, the ASD person can review the plan as often as necessary to understand what to expect. The other use scenario is at the medical center (Fig. 2c), where the ASD person can follow the plan and communicate with the medical professionals via the AACs (they can communicate symptoms, duration, intensity, and body parts affected). PlanTEA supports the following list of functionalities:

- **Anticipation boards**, to plan the activities to be performed before, during, and after the medical appointment (Fig. 3a). This tool allows to break down the steps that make up an activity by means of pictograms and/or images. For example, Fig. 3a represents the pictograms into which the visit to the doctor for a child's growth check can be broken down (*GOING-MEDICAL CENTER-DOCTOR-MEASURE*);
- The available pictograms and images are organized into **categories** (for example, actions, places, persons, types of medical consultations, entertainment during waiting times, and rewards) (Fig. 3a);
- **Drag and drop** the desired pictogram to create the plan;
- **Plan management** allows for their modification, destruction, and storage for future use. (Fig. 3b);
- **Personal photographs** can be uploaded from the mobile device's gallery along with pictograms with the special vocabulary of this area (medical surroundings). This feature makes planning more familiar to the ASD person. For example, a photograph of the medical center usually visited or of the attending physician can be included in the planning instead of the generic pictograms that represent these concepts;
- **Calming objects and entertainment activities** are included to help pass the (expected or unexpected) waiting times during the plan execution by the ASD person (Fig. 4a). The calming object refers to any item that makes the ASD person feel safe, such as a toy, a stuffed animal, or a cartoon, and that is important to have with him/her to decrease the level of anxiety that the situation

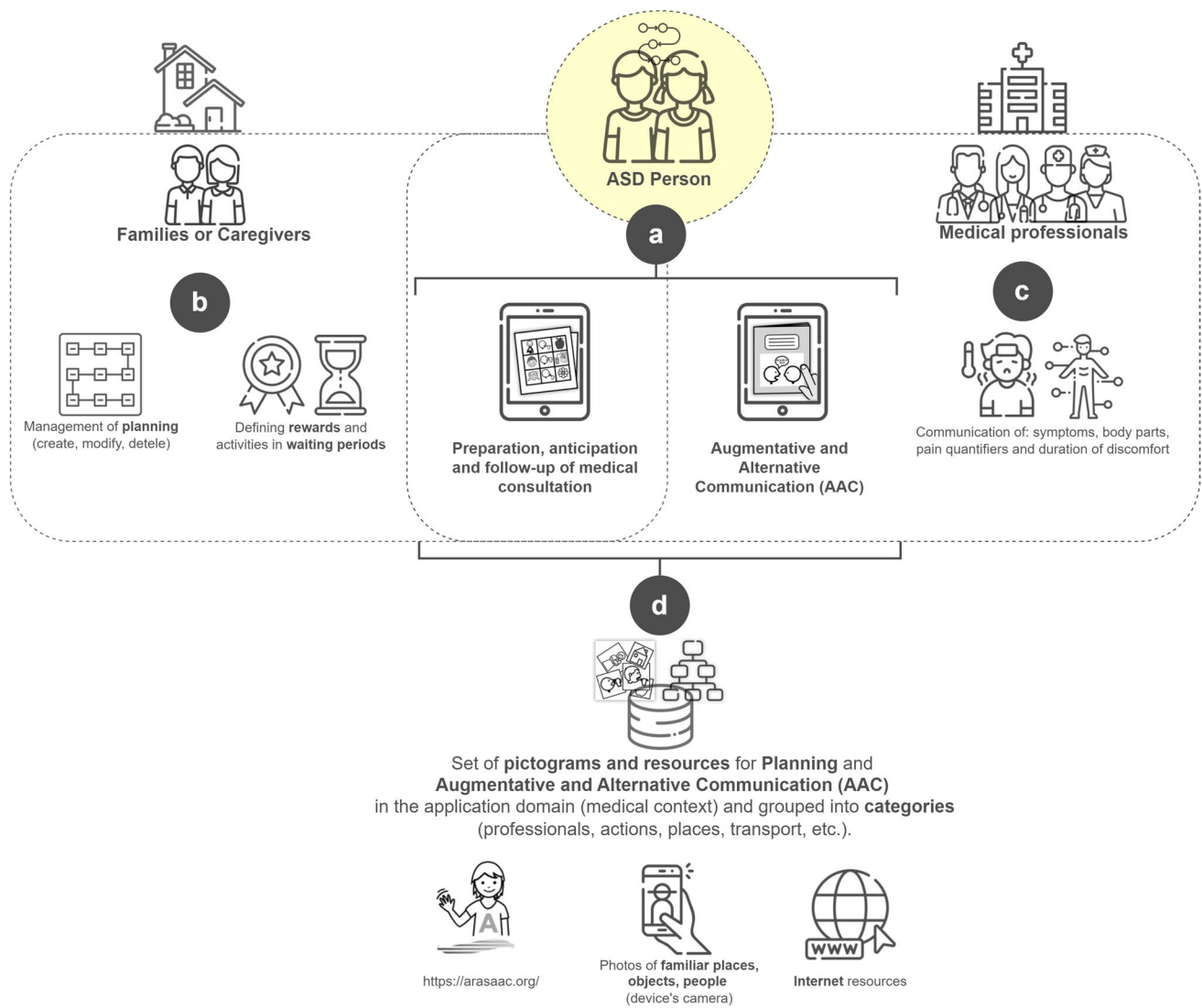


Fig. 2 Overview of PlanTEA

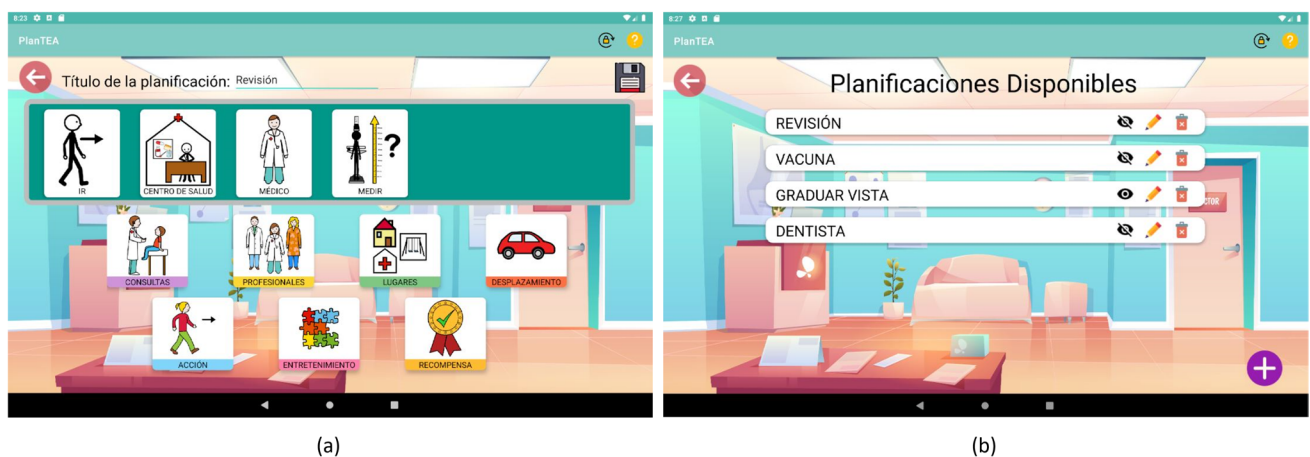


Fig. 3 **a** Planning design (planner role): the plan sequence is generated at the upper side, and the pictogram categories are at the bottom. **b** Management of plans functionality

may cause. Entertainment activities are those that can be used during waiting times, such as watching videos on YouTube, listening to music, reading, etc.;

- **Awards** are provided at specific planning periods (when planners deem it most suitable) (Fig. 4b);
- When necessary, a **communication system** (such as boards for Augmentative and Alternative Communication) between the ASD person and the doctor should be used (non-verbal ASD people or those with oral communication difficulties) (Fig. 5a). This system allows the ASD person to choose from various pictograms that represent signs, symptoms (Fig. 5b), body parts, and the severity and duration of pain or discomfort, among other things.

4 PlanTEA UX evaluation

The UX of PlanTEA has been evaluated using the methodology proposed by Valencia et al. [15]. The description of the process follows the three stages of the methodology.

4.1 Planning stage

First, information on the PlanTEA application was collected, including its characteristics, target users, and limitations:

- *Characteristics:* PlanTEA is a mobile application designed to facilitate the planning of medical appointments for children and adults with ASD and communication with medical specialists. It requires the creation of two profiles: a planner and a user. The planner schedules

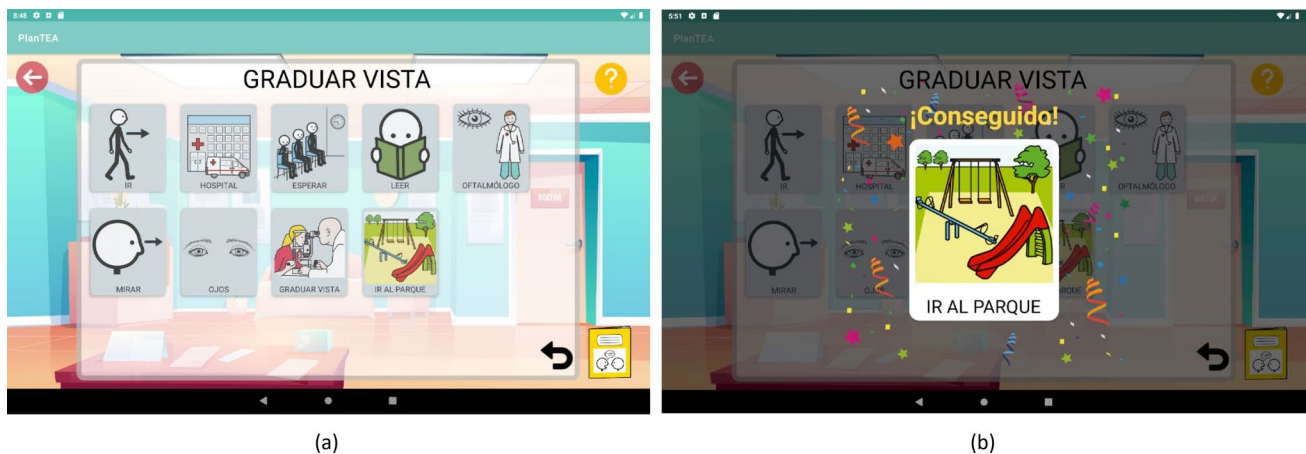


Fig. 4 **a** Example of a plan including an entertainment activity (highlighted with a red border) and a prize (highlighted with a yellow border), **b** appearance of a successful award

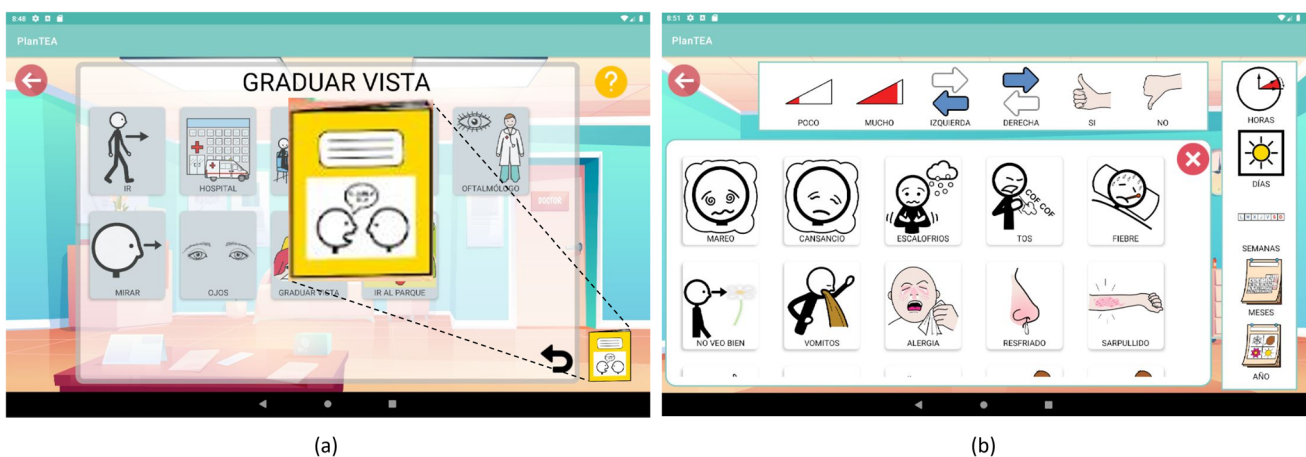
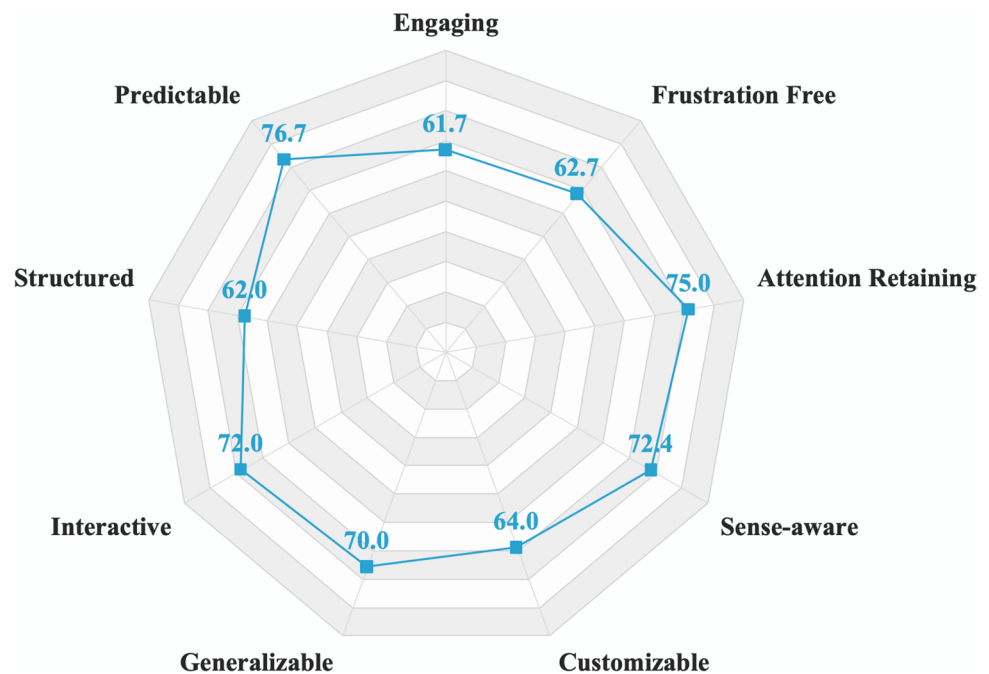


Fig. 5 **a** During the plan execution, the communication board is always accessible. **b** Part of the communication booklet allows the selection of symptoms

Fig. 6 PlanTEA satisfaction percentages



appointments using pictograms that outline the steps before the meeting. Conversely, the user follows these steps and can interact with a calming object and a communication notebook;

- **Target Users:** Children and adults with level 1 ASD who need assistance in planning their medical appointments. Considering the dependence and autonomy of the participants, the support of tutors who can help the participants with problems or questions is needed;
- **Limitations:** PlanTEA is explicitly designed for tablets with a resolution of 2560×1800 running on Android OS. It is not responsive and does not support screen rotation. Given the specific characteristics of the target users, the time available to evaluate the application could be much higher.

This information was crucial for executing the various steps in the planning stage described below.

4.1.1 Method execution planning

The purpose and scope of the UX evaluation and the available time and resources were defined to determine and select the evaluation methods:

- **Objective:** To assess adults with level 1 ASD's perceptions of the PlanTEA application;
- **Scope:** Evaluate the app's functionalities, including creating and viewing schedules and interacting with the communication notebook and the calming object;

- **Time and Resources:** In the case of expert evaluators, there is flexibility regarding time and resources for this evaluation, but not for the target users and their assistants;
- **Evaluation methods:** Given the time and resource constraints, the property checklist, heuristic evaluation, and controlled observation are the evaluation methods selected, as recommended in the methodology [15].

4.1.2 Experiment design

After selecting the evaluation methods, the experiment design was created, including:

- The objectives for each evaluation method: The inspection methods (property checklist and heuristic evaluation) are intended to scrutinize every feature of the mobile application, and the controlled observation is specifically devised to evaluate the application's capability to assist the target users so that they have to plan and display a medical appointment, as well as recognize the calming object;
- The tasks to be carried out by participants in the controlled observation: A scenario was created with two main tasks aimed at helping the user get a flu vaccination. These tasks included planning a medical appointment for the flu shot using the planner profile and later identifying a calming object within the user's profile (Appendix A);
- The protocol to execute the evaluation methods which comprises the following documents:

Table 1 Heuristics used in the PlanTEA inspection created by Ignacio Castro [35]

ID	Heuristic name	Heuristic definition
ASDUX1	System status visibility	The system must always keep users informed about the processes and status changes that occur, to be able to give the user the necessary elements to know what will happen and thus lower uncertainty
ASDUX2	Match between the system and the real world	The system must speak the language of the users with familiar concepts, which must be structured, instead of system-oriented terms so that the user can more easily interpret and generalize
ASDUX3	Consistency and standards	Allow users to customize frequent actions as needed by either shortcuts or toolbars, so that actions are not frustrating because they are unnecessary
ASDUX4	Recognition instead of remembering	Minimize the user's memory load by making objects, actions and options of the system, product, or service visible, so that they do not have to be searched for, thus helping to retain attention on the desired task. The user should not have to remember information from one part of the interface to another. The instructions for use of the system must be visible or easily retrievable whenever appropriate
ASDUX5	Aesthetic and minimalist design	The user interface design should not contain information that is irrelevant or rarely needed, as it can distract the user's attention, so that only important or essential information is displayed, making it easier for the user to perform a task
ASDUX6	User control and freedom	Users can make errors in the system, so there must be a clearly marked "emergency exit" to get out of unwanted states, which allows easy recovery from errors avoiding frustration, increasing confidence, and making system operation predictable. system to the user. The system should allow users to move from one section to another and provide the facility to perform activities repetitively
ASDUX7	Error prevention	The system provides confirmation options which contain information about what you want to do before the user commits to an action to avoid the frustration of making a mistake. If users select wrong options, the system should provide alternative options for them to choose from
ASDUX8	Flexibility and efficiency of use	Allow users to customize frequent actions as needed by either shortcuts or toolbars, so that actions are not frustrating because they are unnecessary
ASDUX9	Help users recognize, diagnose, and recover from errors	Error messages must be expressed in a language understandable to the user. The system should provide multimedia demonstrations to give hints to the user when an error occurs, to avoid frustration from not easily understanding why it occurs or how to overcome it
ASDUX10	Help and documentation	Documentation and any help information should be easy to find, focused on the user's task, providing a multimedia demonstration of the tasks to be performed at an easily viewable size
ASDUX11	Customization of interface elements	The system must allow the customization of the interface elements according to the needs, abilities, and preferences of each user. Interface elements should be large enough that users can read and interact with them without difficulty due to their size. That should also allow them to change various settings for system background, font, color, screen size, and others
ASDUX12	Unexpected changes in the system interface	Changes to the interface must be done step by step. Users cannot cope with sudden or drastic changes being made
ASDUX13	System response capacity	The actions carried out by the user should not have latency since they can lose attention from the task they were doing; this can cause the user to forget and get frustrated easily The system responds to user actions without latency, allowing you to focus on the task at hand without getting frustrated
ASDUX14	Track user activities and monitor performance	The system must keep a history of all the activities carried out by the user, time spent, answers provided, results and others, in case you have not paid the necessary attention, the system must allow the user to return to past activities
ASDUX15	Use of multimodalities for communication	Users must have the option of using different devices to interact with the system. In addition, the system must offer several options to communicate, be it text, audio, images, video, etc

Table 1 (continued)

ID	Heuristic name	Heuristic definition
ASDUX16	Structured interface	The system must maintain a design that allows the user to know, as he interacts, where to find the options, it offers because its structure is logical, simple and coherent, always keeping, according to the needs, everything in its place

Table 2 Tasks list for researchers to evaluate the controlled observation

#	Description	Criteria for success	Task accomplishment	Max time	Remarks and comments
1	Plan an appointment for a flu vaccination	Entered with the planner profile correctly Date is selected The vaccination schedule is selected correctly The type of consultation is selected. It is expected to select the option "Vaccination" The medical consultation is saved correctly		8 min	
2	Actions to be taken at the dentist's surgery	The pictograms representing the actions to be performed are correctly identified Each pictogram is identified as completed		1 min	
3	Activity to be carried out while waiting for the medical consultation	The activity to be performed while waiting for the consultation is correctly identified The recess activity is identified as completed		1 min	
4	Calming object	The screen position of the calming object is correctly identified The calming object is identified The calming object is selected		2 min	
5	Consultation with the dentist	The position on the screen of the consultation notebook is correctly identified The pictogram reflecting the reason for going to the dentist is correctly identified A pictogram reflecting the ailment suffered is correctly identified An option is identified to indicate the intensity of the ailment The intensity of the pain being suffered is correctly indicated A pictogram reflecting the position of the pain suffered is correctly indicated A pictogram reflecting the fact that you have been in pain for 3 days is correctly marked		5 min	
6	Leaving the consultation	The pictograms representing the actions to be performed are correctly identified Each of the pictograms is identified as completed The last activity to be carried out after the consultation with the dentist is correctly identified		1 min	

- o *Confidentiality Agreement:* Every participant in the controlled observation was given a confidentiality agreement according to the Social Research Ethics Committee (SREC) ethical standards of the University of Castilla-La Mancha (ref. CEIS-648868-S1JT). The agreement ensures participant informa-

- tion won't be disclosed, and recordings will be made without capturing faces or voices;
- o *Preliminary Questionnaire and Perception Questionnaire:* The inspection and controlled observation methods were accompanied by preliminary and perception questionnaires. Those for researchers are detailed in Appendix B.1 and B.2, and those for the participants are shown in Appendix B.3 and B.4.

The latter were previously adapted to the comprehension needs of each participant;

- o *Property Checklist for Expert Evaluators*: This checklist comprised nine categories and 26 items, as seen in Appendix C, and was specifically tailored to assess the UX for people with level 1 ASD;
- o *Heuristics List for Expert Evaluators*: A set of 16 heuristics was explicitly designed for systems used by people with ASD [35], as Table 1 in Appendix D illustrates;
- o *Observer logs*: During the controlled observation, the researchers would record the interactions and behavior of each participant with the PlanTEA application;
- o *Tasks list*: Two task lists were distributed to the participants and the researchers, respectively. The participants' task list is presented in Appendix A, and Table 2 of Appendix E shows the researchers' one, including detailed information about the expected results and times.

4.1.3 Evaluators selection

The study involved ten evaluators (five for each inspection method), who were contacted by e-mail. The experts' profiles include Ph.D. students, computer scientists with UX/usability expertise, and domain-specific experts, such as psychologists and counselors with practical experience working with people with ASD. Two evaluators were professionals and workers of the Regional Association of People Affected by Autism and Other Developmental Disorders (AUTRADE¹).

4.1.4 Participants selection

A controlled observation was carried out with seven individuals diagnosed with ASD level 1, facilitated by the AUTRADE association. Two professionals from AUTRADE also participated as tutors to support the participants.

4.2 Execution stage

Before evaluating PlanTEA, evaluators were briefed about the app and its purpose. They were also provided with a glossary explaining concepts of *user experience*, *autism spectrum disorder*, and the *characteristics of ASD people*. Information about the evaluation process was also given. The instructions for the evaluation were sent to the evaluators via email, as well as an Excel file containing instructions, preliminary and perception questionnaires (Appendix B.1 and B.2, respectively), and formatted sheets to fill out

depending on the inspection method to be used. Due to the available time, the evaluation methods were carried out in a specific sequence: controlled observation was conducted first, followed by the property checklist and heuristic evaluation performed concurrently.

4.2.1 Preliminary evaluation

Five expert evaluators with knowledge in UX and ASD freely interacted with the PlanTEA app and assessed each item of the property checklist (Appendix C) on a scale from 1 (not complied at all) to 5 (fully complied), with the option to add comments or observations for each item, and optionally provide comments/observations on each item.

4.2.2 Inspections

For the heuristic evaluation, the other five experts received a detailed list of the selected heuristics (Appendix D) and a spreadsheet to be completed by the evaluator with each potential problem found after interacting with the PlanTEA application. They were asked to find at least ten problems, providing a definition, explanation, the path taken to discover the problem, an image example, and the unmet heuristic for each problem. All issues identified were consolidated into a single list, merging duplicates. Once all the potential problems identified by the evaluators were compiled and consolidated into a single list, the evaluators were asked to assign a frequency and severity score for each potential problem.

4.2.3 User tests

The PlanTEA user tests were held at AUTRADE's installations and were split into two sessions, with three users participating in the first session and four in the second. The process was supported by a team of four experts, of which two were UX and ASD experts, and the other two were professionals from AUTRADE. Everything was carefully designed to ensure that the ASD users felt comfortable and safe throughout the process. First, participants were contacted before the assessment day. The researchers received guidelines from AUTRADE's ASD professionals about each participant's characteristics. This information was used to tailor the evaluation process to each participant.

Before starting the evaluation tasks, participants were provided textual instructions about the experiment. They were also asked to read, consent, and respond to a confidentiality agreement and preliminary questionnaire. Participants were guided to an office equipped for the assessment process during the evaluation. After interacting with the PlanTEA application, participants were asked to complete a perception questionnaire about their experience with the application.

¹ <http://autrade.info>.

Table 3 Potential PlanTEA problems encountered by evaluators

ID	Problem definition	Comments and explanation	Where	Unfulfilled heuristic
P1	There is no differentiation between users and objects	The iconography and way of entering the info for the planner/user and "reassuring object" users is similar and may suggest that the "reassuring object" may be another user/actor of the system	Preferences	ASDUX2
P2	Options use iconography and do not include text that represents	Most of the icons used in the system do not have a text that shows what action they perform. This is often confusing	Preferences	ASDUX15
P3	It is not allowed to "schedule" without selecting the default time	The "schedule" button is disabled until a time is selected, not allowing the default time to be selected. This also applies to select the "consultation type"	Planner	ASDUX8
P4	The "schedule" button does not schedule immediately	When you press plan, you must then press "new planning" to start the process. The middle step seems unnecessary and misleads from expectations	Planner	ASDUX13
P5	Schedule date and time are not displayed throughout the process	When starting a schedule, the time and date selected on the previous screen are not displayed	Planner	ASDUX4
P6	Difficult pictogram movement in planning	When selecting schedule cards, they must be dragged "hard" to be able to move them	Planner	ASDUX13
P7	Confusing pictogram categories	It is not clear which pictograms are "categories" that include more pictograms inside, since they share a similar design with the pictograms that are selectable	Planner	ASDUX3
P8	Difficulty removing a pictogram	The action to "remove" a glyph from the timeline is not intuitive. To remove a card, you must drag it up, which is the same action used to move from the bottom selection to the timeline	Planner	ASDUX3
P9	Step repetition to select pictograms from the "action" category	The pictograms in the "action" category are probably the most frequently used in the system, so it is inefficient to go back through the menus to select them	Planner	ASDUX8
P10	Frame consistency for different types of pictograms	The Pictograms have different frames to represent the "entertainment" and "reward" categories, while the rest have no frame at all. This can be distracting and is inconsistent	Planner	ASDUX5
P11	Hard to see error messages and confirmation of actions	The error message when trying to save without entering a title when creating a schedule is small and inconspicuous, which may cause the issue to be ignored and not easily resolved	Planner	ASDUX7
P12	Activity and event icons are unclear	Activity icons aren't clear, and it's easy to accidentally create duplicate events by not forcing the title change	Planner	ASDUX9
P13	Non-modifiable consultation type and time	The events cannot be changed of day and/or type of planning	Planner	ASDUX8
P14	Elimination of progress when changing day	Changing the day after creating an event without pressing save on the screen causes the progress to be deleted. There is no warning about it	Planner	ASDUX9
P15	Confusing and inconsistent event visibility	When you select the day again and press schedule, the event appears again The visibility icon of an event is not clear regarding the status of the event. In addition, only one event is allowed to be visible, despite being on different days	Planner	ASDUX4

Table 3 (continued)

ID	Problem definition	Comments and explanation	Where	Unfulfilled heuristic
P16	Events visible out of date	Events visible in the user's profile even though said event was scheduled for previous days. Example: The user can interact with the event scheduled for October 15 and at the time of review this is November 2	Planner	ASDUX3
P17	Difficulty and inconsistency in saving an event on a specific day	To be able to save an event created in a day, it must be selected by holding down the card and then pressing save. This is very unintuitive when it comes to assigning the event to the day it was created	Planner	ASDUX6
P18	Scroll not evident in communication notebook box	The box that contains the pictograms in the user's communication notebook does not seem to be scrollable (move), although it is. It is possible not to see that there are more pictograms below those shown by default	Communications notebook	ASDUX1
P19	Unclear and contradictory pictogram intensity	The pictograms in the communication notebook show a bar indicating "intensity", it is not immediately clear what it represents. In addition, the pictograms are presented with a negative connotation, so it is possible that selecting the intensity can cause a double negation of the user's intention	Communications notebook	ASDUX2
P20	Difficulty closing pictograms	The pictograms displayed on the plan are difficult to close in the app, as there is an area around the image where the image cannot be closed. There is also no close icon or button	Planner	ASDUX1
P21	Inconsistency and animations in entertainment pictograms and rewards	Pictograms for entertainment and rewards have different frames and play animations. These animations only apply to this type of pictograms and can be "surprising"	Planner	ASDUX12
P22	Unintuitive user manual	The user manual is difficult to navigate and is not consistent in its section numbering. Section 2 includes subsections shown with dashes; however, the others include subsections with subpoints like 4	User Manual	ASDUX10
P23	Application responsiveness on different devices	The application is not adapted to all types of tablet screens. This produces that some texts or sections are not displayed correctly or are difficult to use	General	ASDUX7
P24	Too much information	By pressing the "user manual" button, the system always offers the same manual with complete information and must be focused on the user's task	General	ASDUX10
P25	No customization of interface elements	The system does not allow customizing the interface elements so that they are adapted to the user's needs	General	ASDUX11
P26	Lack of audio help in the user manual	The application lacks audio aids to read/explain sections of the user manual	User Manual	ASDUX15
P27	Actions carried out with unrepresentative icon	Marking actions performed in gray may not be so noticeable for the user to perceive this task as done, so there should be another way to represent when an action is complete	Planner	ASDUX11

Table 3 (continued)

ID	Problem definition	Comments and explanation	Where	Unfulfilled heuristic
P28	Confirm a schedule	When pressing the "Save" button, the system must ask to confirm this option to avoid errors. I understand that this section of the system is not intended for the user with ASD, but I think that the user can participate in the entire process (creation and monitoring of events)	Planner	ASDUX7
P29	Make actions easier	It is possible that being a long planning, this is frustrating for the user having to drag all the actions/pictogram. The system should allow adding tasks by pressing the action/pictogram without dragging it	Planner	ASDUX8
P30	Unexpected closing of the app	When you click on the arrow (upper left) of the application, it closes unexpectedly	Preferences	ASDUX7
P31	Unnecessary information	Information appears as blocked, instead of appearing only when the "appointment notifications" option is selected	Preferences	ASDUX5
P32	User manual slow to load	Clicking on the user manual does not load immediately. A loading screen should be provided indicating that it was pressed and that the button is blocked so that it cannot be clicked further	Preferences	ASDUX1
P33	Lack of welcome tour	If I hadn't been curious about the documentation, I wouldn't have found out that I was making settings for the first-time use of the application, a tour is needed showing each functionality and what each thing means	Preferences	ASDUX10
P34	Lack of linguistic connectors	Message with missing linguistic connectors. What can cause that at first instance the "planner information" section is not understood. This can be interpreted as another object or a person/profile	Preferences	ASDUX3
P35	Password modification icon	The planner profile password change icon is unclear	Preferences	ASDUX2
P36	Difficulty closing popup screen	Inside the red box there is a layer that does not allow you to exit the page, if you press outside the red box the pictogram closes but otherwise it does not answer or answers with a delay	Communications notebook	ASDUX13
P37	Delete an event	When you delete an event, it doesn't immediately disappear from the calendar	Planner	ASDUX2
P38	Image of profiles and calming object	The application only allows uploading an image from the device's gallery. Cannot access the camera to take a photo at the moment	Preferences	ASDUX2
P39	Obligation of actions of different types of users	There are multiple elements and actions of the application that are mandatory, however these are not necessary for some user profiles that will use the application (for example: adults who can create their own schedules), as are the two types of profiles, passwords, and the use of a reassuring object	Preferences	ASDUX8
P40	Non-descriptive consultation type	Some of the options provided by "Type of consultation" are not very descriptive, which may confuse the user when he/she wants to plan his/her medical consultation. Example: "Tests" option	Planner	ASDUX3
P41	Confusion between the "New planning" button and the planning section	When it says new planning, it seems that what you must do is another plan, not that it is within the same plan	Planner	ASDUX3

Table 3 (continued)

ID	Problem definition	Comments and explanation	Where	Unfulfilled heuristic
P42	User profile has no calendar	In case the planner is also the user, the application should present the same calendar shown in the planner's profile	General	ASDUX16
P43	Intention of the communications notebook	The purpose and intention of the communications notebook is not clear since a description of its use is missing	Communications notebook	ASDUX10

4.3 Results analysis stage

The results obtained during the evaluations were organized and analyzed as described in the following sub-sections.

4.3.1 Grouping of potential problems

Each evaluator identified 10 to 21 potential problems. These were merged to remove duplicates to obtain the 43 problems in Table 3 (see Appendix F). The evaluators' expertise in UX/Usability evaluations and their knowledge of the characteristics of people with ASD are believed to contribute to the diversity of issues found. Then, heuristics were associated with the potential problems found, as seen in Table 4 (Appendix G). Most are related to heuristics related to the matching between the system and the real world (ASDUX2), consistency and standards (ASDUX3), and aesthetic and minimalist design (ASDUX8), with 6, 5, and 5 potential problems, respectively. This may be because the expectations of a person with ASD regarding real-world alignment, standards, and flexibility when using PlanTEA are determined by their previous interaction with technological systems. Interestingly, no potential problem was linked to the heuristic related to tracking user activities and monitoring performance (ASDUX14). It could be because the characteristics and objectives of PlanTEA do not involve activities that need supervision, which makes this heuristic inapplicable to the system's current state.

On the other hand, Table 5 (Appendix H) shows the values given by each evaluator for the frequency, severity, and criticality of each problem, illustrating how much importance each evaluator provides to it. There are eight potential problems (P4, P9, P13, P22, P33, P34, P42, and P43) with standard deviations greater than 2.06, which suggests a divergence in the evaluators' views regarding the ASD domain. These differences emphasize the importance of having experts from different knowledge areas (UX, ASD, and both).

When sorting problems by criticality, Table 6 (Appendix I) shows that five problems exceeded 7.0 on a scale of 0–8. These are P2 (iconography options lacking text representation), P17 (difficulty and inconsistency in saving an event on a specific day), P6 (difficulty moving pictograms in planning), P14 (progress deletion when changing a day), and P16 (events visible outside the date). Therefore, these problems should be addressed first to improve user experience with PlanTEA. According to severity, Table 7 (Appendix J) shows that eighteen issues had a severity score above 3.0 on a 0–4 scale, with five problems scoring above 3.6. These include P6 (difficulty moving pictograms in planning), P14 (progress deletion when changing a day), P16 (events visible outside the date), and P17 (difficulty and inconsistency in saving

Table 4 Unfulfilled heuristics by the potential PlanTEA problems

ID	Heuristic	Associated potential problem(s)	Number of problems
ASDUX1	System state visibility	P18, P20, P32	3
ASDUX2	Match between the system and the real world	P1, P19, P35, P37, P38	5
ASDUX3	Consistency and standards	P7, P8, P16, P34, P40, P41	6
ASDUX4	Recognition instead of remembering	P5, P15	2
ASDUX5	Aesthetic and minimalist design	P10, P31	2
ASDUX6	User control and freedom	P17	1
ASDUX7	Error prevention	P11, P23, P28, P30	4
ASDUX8	Flexibility and efficiency of use	P3, P9, P13, P29, P39	5
ASDUX9	Help users recognize, diagnose, and recover from errors	P12, P14	2
ASDUX10	Help and documentation	P22, P24, P33, P43	4
ASDUX11	Customization of interface elements	P25, P27	2
ASDUX12	Unexpected system interface changes	P21	1
ASDUX13	System Responsiveness	P4, P6, P36	3
ASDUX14	Track user activities and monitor performance		0
ASDUX15	Use of multimodalities for communication	P2, P26	2
ASDUX16	Structured interface	P42	1

an event on a specific day), among the most critical, and P8 (difficulty deleting a pictogram).

Upon comparing the potential problems with higher criticality and severity, it was observed that:

- 80% of the top five critical problems are also part of the top five severe problems (P17, P6, P14, and P16);
- Although problem P2 (iconography options lacking text representation) is considered the most critical, it ranks eighth in severity due to its maximum frequency score of 4;
- Problem P8 (difficulty removing a pictogram) is fourth in severity but eighth in criticality because it has a frequency score of 2.8 out of a maximum of 4.

4.3.2 Quantitative analysis

The data gathered from the conducted evaluations was analyzed quantitatively and organized in the following way:

- *Preliminary Demographic Questionnaire for Evaluators:* The group of experts is quite diverse, comprising Ph.D. students, computer scientists specializing in UX/Usability, and professionals like psychologists and counselors with direct experience with individuals with ASD. Some of these experts have expertise in both UX/Usability and ASD, and some are themselves diagnosed with ASD. In fact, 90% of the experts were already familiar with the concepts of UX/Usability, while all of them were

acquainted with the ASD concept. Furthermore, 30% of the experts had prior experience using applications or systems designed specifically for people with ASD;

- *Preliminary Demographic Questionnaire for Participants:* The group consisted of two females and five males, aged between 20 and 26, with an average age of 22.57. It was found that 57.14% (4 participants) rarely plan their medical consultations. The same percentage found it occasionally easy to communicate with doctors during consultations and felt satisfaction afterward. A large majority, 85.71%, use devices such as tablets, mobile phones, or computers daily. However, the same percentage reported never using applications specifically designed for people with ASD;
- *Perception Questionnaire for Evaluators:* While 60% of the evaluators found the application helpful during interaction, 70% found it difficult to interact with. Half of the evaluators felt their interactions were ineffective and were dissatisfied with the app. There needed to be a consensus on recommending the application for scheduling medical appointments and communicating with specialists. Half of the evaluators found the application appealing to ASD users, but there were doubts about its ability to foster a sense of trust. The evaluators agreed that the application elements are ordered and structured coherently, resembling those in real life. The application was considered customizable by half of the evaluators and designed with ASD users' sensory characteristics in mind. Most evaluators agreed that the app keeps the attention of ASD users and manages time adequately, but

Table 5 Frequency, Severity, and Criticality values assigned by evaluators to potential PlanTEA problems

ID	Problem definition	Ev #1			Ev #2			Ev #3			Ev #4			Ev #5			Mean			Standard deviation		
		S	F	C	S	F	C	S	F	C	S	F	C	S	F	C	S	F	C	S	F	C
P1	There is no differentiation between users and objects	2	1	3	2	2	4	3	2	5	4	4	8	2	3	5	2.60	2.40	5.00	0.80	1.02	1.67
P2	Options use iconography and do not include text that represents	3	4	7	3	4	7	3	4	7	4	4	8	3	4	7	3.25	4.00	7.25	0.39	0.00	0.39
P3	It is not allowed to "schedule" without selecting the default time	2	2	4	1	3	4	3	1	4	4	4	8	4	4	8	2.80	2.80	5.60	1.17	1.17	1.96
P4	The "schedule" button does not schedule immediately	4	3	7	1	3	4	2	1	3	4	4	8	4	4	8	3.00	3.00	6.00	1.26	1.10	2.10
P5	Schedule date and time are not displayed throughout the process	2	2	4	2	3	5	2	2	4	3	4	7	2	4	6	2.20	3.00	5.20	0.40	0.89	1.17
P6	Difficult pictogram movement in planning	4	3	7	4	3	7	3	3	6	4	3	7	4	4	8	3.80	3.20	7.00	0.40	0.40	0.63
P7	Confusing pictogram categories	2	3	5	4	4	8	3	2	5	4	4	8	4	4	8	3.40	3.40	6.80	0.80	0.80	1.47
P8	Difficulty removing a pictogram	3	3	6	4	3	7	3	2	5	4	3	7	4	3	7	3.60	2.80	6.40	0.49	0.40	0.80
P9	Step repetition to select pictograms from the "action" category	2	2	4	2	3	5	2	1	3	4	4	8	4	4	8	2.80	2.80	5.60	0.98	1.17	2.06
P10	Frame consistency for different types of pictograms	1	2	3	2	3	5	2	3	5	2	2	4	2	3	5	1.80	2.60	4.40	0.40	0.49	0.80
P11	Hard to see error messages and confirmation of actions	2	3	5	2	2	4	1	2	3	3	2	5	4	4	8	2.40	2.60	5.00	1.02	0.80	1.67
P12	Activity and event icons are unclear	2	2	4	1	2	3	2	2	4	4	2	6	2	2	4	2.20	2.00	4.20	0.98	0.00	0.98
P13	Non-modifiable consultation type and time	1	2	3	1	2	3	3	3	6	4	4	8	4	4	8	2.60	3.00	5.60	1.36	0.89	2.24
P14	Elimination of progress when changing day	4	2	6	4	3	7	3	3	6	4	4	8	4	4	8	3.80	3.20	7.00	0.40	0.75	0.89
P15	Confusing and inconsistent event visibility	3	3	6	2	3	5	3	3	6	4	4	8	3	2	5	3.00	3.00	6.00	0.63	0.63	1.10
P16	Events visible out of date	3	2	5	4	3	7	4	3	7	4	4	8	4	4	8	3.80	3.20	7.00	0.40	0.75	1.10
P17	Difficulty and inconsistency in saving an event on a specific day	4	3	7	3	4	7	3	3	6	4	4	8	4	4	8	3.60	3.60	7.20	0.49	0.49	0.75
P18	Scroll not evident in communication notebook box	2	2	4	2	4	6	3	3	6	4	4	8	4	4	8	3.00	3.40	6.40	0.89	0.80	1.50
P19	Unclear and contradictory pictogram intensity	2	2	4	2	4	6	2	3	5	4	4	8	3	3	6	2.50	3.25	5.75	0.77	0.74	1.32
P20	Difficulty closing pictograms	2	2	4	2	3	5	3	2	5	4	3	7	3	3	6	2.80	2.60	5.40	0.75	0.49	1.02
P21	Inconsistency and animations in entertainment pictograms and rewards	2	2	4	1	3	4	3	1	4	3	2	5	2	2	4	2.20	2.00	4.20	0.75	0.63	0.40
P22	Unintuitive user manual	3	2	5	1	1	2	3	1	4	4	4	8	4	4	8	3.00	2.40	5.40	1.10	1.36	2.33
P23	Application responsiveness on different devices	4	3	7	3	4	7	4	2	6	2	2	4	1	1	2	2.80	2.40	5.20	1.17	1.02	1.94
P24	Too much information	3	3	6	2	4	6	3	2	5	4	4	8	4	4	8	3.20	3.40	6.60	0.75	0.80	1.20
P25	No customization of interface elements	2	3	5	2	4	6	2	2	4	4	4	8	4	4	8	2.80	3.40	6.20	0.98	0.80	1.60
P26	Lack of audio help in the user manual	2	2	4	2	4	6	3	2	5	4	4	8	4	4	8	3.00	3.20	6.20	0.89	0.98	1.60
P27	Actions carried out with unrepresentative icon	2	2	4	2	3	5	3	2	5	4	4	8	2	2	4	2.60	2.60	5.20	0.80	0.80	1.47
P28	Confirm a schedule	3	2	5	3	4	7	2	2	4	4	4	8	4	4	8	3.20	3.20	6.40	0.75	0.98	1.62
P29	Make actions easier	2	2	4	3	4	7	2	2	4	4	4	8	4	4	8	3.00	3.20	6.20	0.89	0.98	1.83
P30	Unexpected closing of the app	4	1	5	2	1	3	3	3	6	4	4	8	4	4	8	3.40	2.60	6.00	0.80	1.36	1.90
P31	Unnecessary information	2	2	4	2	2	4	2	2	4	4	3	7	3	3	6	2.60	2.40	5.00	0.80	0.49	1.26
P32	User manual slow to load	2	2	4	1	1	2	2	2	4	2	2	4	4	4	8	2.20	2.20	4.40	0.98	0.98	1.96
P33	Lack of welcome tour	3	1	4	0	1	1	4	1	5	4	4	8	4	4	8	3.00	2.20	5.20	1.55	1.47	2.64
P34	Lack of linguistic connectors	1	3	4	1	1	2	3	3	6	4	4	8	4	4	8	2.60	3.00	5.60	1.36	1.10	2.33

Table 5 (continued)

ID	Problem definition	Ev #1			Ev #2			Ev #3			Ev #4			Ev #5			Mean			Standard deviation		
		S	F	C	S	F	C	S	F	C	S	F	C	S	F	C	S	F	C	S	F	C
P35	Password modification icon	1	2	3	1	1	2	2	1	3	2	2	4	2	2	4	1.60	1.60	3.20	0.49	0.49	0.75
P36	Difficulty closing popup screen	1	2	3	2	3	5	3	2	5	4	4	8	2	2	4	2.40	2.60	5.00	1.02	0.80	1.67
P37	Delete an event	2	2	4	2	4	6	2	3	5	2	2	4	2	2	4	2.00	2.60	4.60	0.00	0.80	0.80
P38	Image of profiles and calming object	1	1	2	1	2	3	2	2	4	3	4	7	2	2	4	1.80	2.20	4.00	0.75	0.98	1.67
P39	Obligation of actions of different types of users	2	3	5	1	2	3	3	2	5	4	4	8	3	3	5	2.50	2.75	5.25	1.00	0.74	1.60
P40	Non-descriptive consultation type	2	2	4	1	3	4	3	2	5	3	3	6	4	4	8	2.60	2.80	5.40	1.02	0.75	1.50
P41	Confusion between the "New planning" button and the planning section	2	2	4	3	4	7	3	2	5	4	4	8	4	4	8	3.20	3.20	6.40	0.75	0.98	1.62
P42	User profile has no calendar	2	2	4	0	2	2	2	2	4	4	4	8	4	4	8	2.40	2.80	5.20	1.50	0.98	2.40
P43	Intention of the communications notebook	1	2	3	0	1	1	3	2	5	4	4	8	4	4	8	2.40	2.60	5.00	1.62	1.20	2.76

there was substantial disagreement on whether it avoids causing frustration during interaction. The use of pictograms and a communication notebook was praised, but there was a demand for a tutorial, more customization options, and better planning management;

- *Perception Questionnaire for Participants:* Approximately 57.14% of the participants found the task completion and information finding to be of neutral difficulty. However, 42.85% felt that the app needed to provide more details for task execution. The understandability of the information provided was found to be neutral by 42.85% of participants, while 28.57% found it could have been more understandable. Regarding guidance, 42.85% had a neutral opinion, while an equal percentage felt fully guided when planning medical consultations. A significant 71.42% found the application usage challenging. Opinions were divided on whether the information was overwhelming and whether the application was customizable. The visual and sound elements of the application did not bother 42.85% of participants, and 28.57% felt they needed to be more focused while interacting with it. About 42.85% were neutral about whether the application helped them rectify errors, and 28.57% would use PlanTEA again for planning medical visits. Participants liked PlanTEA's features that allowed for better self-organization, independent medical consultation planning, and the use of pictograms. However, they disliked the delayed response to their actions and difficulty interacting with elements and found some elements non-intuitive and hard to visualize;
- *Property checklist:* The satisfaction percentages for each category are summarized in Fig. 6.

The categories of predictability and attention retention scored the highest, with satisfaction percentages of 76.7% and 75%, respectively. Regarding predictability, evaluators found the application's behavior coherent and free of arbitrary responses. However, they noted room for improvement in repetitive actions and found some animations and interactions unpredictable. Concerning attention-retaining, the application generally worked correctly, with only minor delays. There were no distracting elements, but some evaluators felt the app sometimes had excess buttons and information. The lowest satisfaction percentages were in the engaging (61.7%), structured (62%), and frustration-free (62.7%) categories. In the first case, evaluators noted a lack of feedback and a history of actions, although they appreciated the visual stimuli for correct actions.

On the other hand, evaluators found inconsistencies in the application's structure, particularly in the event creation process, but noted that navigation was generally consistent. Finally, regarding frustration, evaluators stated a lack of error recovery options, issues with

Table 6 Ranking of potential problems on PlanTEA by criticality

ID	Problem definition	Average		
		Severity	Frequency	Criticality
P2	Options use iconography and do not include text that represents	3.25	4.00	7.25
P17	Difficulty and inconsistency in saving an event on a specific day	3.60	3.60	7.20
P6	Difficult pictogram movement in planning	3.80	3.20	7.00
P14	Elimination of progress when changing day	3.80	3.20	7.00
P16	Events visible out of date	3.80	3.20	7.00
P7	Confusing pictogram categories	3.40	3.40	6.80
P24	Too much information	3.20	3.40	6.60
P8	Difficulty removing a pictogram	3.60	2.80	6.40
P18	Scroll not evident in communication notebook box	3.00	3.40	6.40
P28	Confirm a schedule	3.20	3.20	6.40
P41	Confusion between the "New planning" button and the planning section	3.20	3.20	6.40
P25	No customization of interface elements	2.80	3.40	6.20
P26	Lack of audio help in the user manual	3.00	3.20	6.20
P29	Make actions easier	3.00	3.20	6.20
P4	The "schedule" button does not schedule immediately	3.00	3.00	6.00
P15	Confusing and inconsistent event visibility	3.00	3.00	6.00
P30	Unexpected closing of the app	3.40	2.60	6.00
P19	Unclear and contradictory pictogram intensity	2.50	3.25	5.75
P3	It is not allowed to "schedule" without selecting the default time	2.80	2.80	5.60
P9	Step repetition to select pictograms from the "action" category	2.80	2.80	5.60
P13	Non-modifiable consultation type and time	2.60	3.00	5.60
P34	Lack of linguistic connectors	2.60	3.00	5.60
P20	Difficulty closing pictograms	2.80	2.60	5.40
P22	Unintuitive user manual	3.00	2.40	5.40
P40	Non-descriptive consultation type	2.60	2.80	5.40
P39	Obligation of actions of different types of users	2.50	2.75	5.25
P5	Schedule date and time are not displayed throughout the process	2.20	3.00	5.20
P23	Application responsiveness on different devices	2.80	2.40	5.20
P27	Actions carried out with unrepresentative icon	2.60	2.60	5.20
P33	Lack of welcome tour	3.00	2.20	5.20
P42	User profile has no calendar	2.40	2.80	5.20
P1	There is no differentiation between users and objects	2.60	2.40	5.00
P11	Hard to see error messages and confirmation of actions	2.40	2.60	5.00
P31	Unnecessary information	2.60	2.40	5.00
P36	Difficulty closing popup screen	2.40	2.60	5.00
P43	Intention of the communications notebook	2.40	2.60	5.00
P37	Delete an event	2.00	2.60	4.60
P10	Frame consistency for different types of pictograms	1.80	2.60	4.40
P32	User manual slow to load	2.20	2.20	4.40
P12	Activity and event icons are unclear	2.20	2.00	4.20
P21	Inconsistency and animations in entertainment pictograms and rewards	2.20	2.00	4.20
P38	Image of profiles and calming object	1.80	2.20	4.00
P35	Password modification icon	1.60	1.60	3.20

the visibility of error messages, and the complexity of documentation despite finding it helpful. In summary, all categories scored above 61% in terms of satisfaction. The application's predictability and ability to retain attention

were positively highlighted. However, there is a need for clear feedback, consistency in elements, and better error handling to improve user experience;

Table 7 Ranking of Potential Problems on PlanTEA by severity

ID	Problem definition	Average		
		Severity	Frequency	Criticality
P6	Difficult pictogram movement in planning	3.80	3.20	7.00
P14	Elimination of progress when changing day	3.80	3.20	7.00
P16	Events visible out of date	3.80	3.20	7.00
P8	Difficulty removing a pictogram	3.60	2.80	6.40
P17	Difficulty and inconsistency in saving an event on a specific day	3.60	3.60	7.20
P7	Confusing pictogram categories	3.40	3.40	6.80
P30	Unexpected closing of the app	3.40	2.60	6.00
P2	Options use iconography and do not include text that represents	3.25	4.00	7.25
P24	Too much information	3.20	3.40	6.60
P28	Confirm a schedule	3.20	3.20	6.40
P41	Confusion between the "New planning" button and the planning section	3.20	3.20	6.40
P4	The "schedule" button does not schedule immediately	3.00	3.00	6.00
P15	Confusing and inconsistent event visibility	3.00	3.00	6.00
P18	Scroll not evident in communication notebook box	3.00	3.40	6.40
P22	Unintuitive user manual	3.00	2.40	5.40
P26	Lack of audio help in the user manual	3.00	3.20	6.20
P29	Make actions easier	3.00	3.20	6.20
P33	Lack of welcome tour	3.00	2.20	5.20
P3	It is not allowed to "schedule" without selecting the default time	2.80	2.80	5.60
P9	Step repetition to select pictograms from the "action" category	2.80	2.80	5.60
P20	Difficulty closing pictograms	2.80	2.60	5.40
P23	Application responsiveness on different devices	2.80	2.40	5.20
P25	No customization of interface elements	2.80	3.40	6.20
P1	There is no differentiation between users and objects	2.60	2.40	5.00
P13	Non-modifiable consultation type and time	2.60	3.00	5.60
P27	Actions carried out with unrepresentative icon	2.60	2.60	5.20
P31	Unnecessary information	2.60	2.40	5.00
P34	Lack of linguistic connectors	2.60	3.00	5.60
P40	Non-descriptive consultation type	2.60	2.80	5.40
P19	Unclear and contradictory pictogram intensity	2.50	3.25	5.75
P39	Obligation of actions of different types of users	2.50	2.75	5.25
P11	Hard to see error messages and confirmation of actions	2.40	2.60	5.00
P36	Difficulty closing popup screen	2.40	2.60	5.00
P42	User profile has no calendar	2.40	2.80	5.20
P43	Intention of the communications notebook	2.40	2.60	5.00
P5	Schedule date and time are not displayed throughout the process	2.20	3.00	5.20
P12	Activity and event icons are unclear	2.20	2.00	4.20
P21	Inconsistency and animations in entertainment pictograms and rewards	2.20	2.00	4.20
P32	User manual slow to load	2.20	2.20	4.40
P37	Delete an event	2.00	2.60	4.60
P10	Frame consistency for different types of pictograms	1.80	2.60	4.40
P38	Image of profiles and calming object	1.80	2.20	4.00
P35	Password modification icon	1.60	1.60	3.20

- *List of tasks:* For the two tasks planned in the controlled observation, 8 and 2 min were estimated for the first and second tasks, respectively. The results show that the expected results were underestimated, as the participants

averaged about 18 min. However, the time given to the participants helped them to stay calm and relaxed.

4.3.3 Qualitative analysis

The qualitative evaluations provided the following findings:

- Regarding participants, the *tasks list* and the *observers' logs* were merged for each one and supplemented the recorded videos of the experience of each one, obtaining the following reports:
- **P#1:** The user needed help interacting with the pictograms while planning events in the user profile. The user confused the “right” pictogram with the action of going back and had trouble adding, removing, or changing the order of pictograms. The user successfully found the calming object but lost focus due to discomfort with the pictograms. The tasks took them approximately 31 min to complete;
- **P#2:** The user needed help using the tablet and selecting a category during planning. The user was also frustrated that the application did not save their information when returning to the tabs. The user managed to add only one pictogram to the created event sequence. The user identified the calming object with difficulty and mentioned that the application was “inaccurate”. The tasks took them approximately 14 min to complete;
- **P#3:** The user believed they needed to click and not drag to interact with the pictograms. The user faced challenges in modifying pictograms during event planning and distinguishing between a category or a pictogram for the event sequence. The “back” buttons confused the user during planning, but the calming object was successfully found. The tasks took them approximately 17 min to complete;
- **P#4:** The user needed help finding the option to add a new event and interacting with the pictograms when planning an event. The user thought they could access the user profile from the top option in the planner profile, which has a “user” icon. The user mentioned that the calming objects were the ball and the communication notebook but did not interact with them appropriately. The tasks took them approximately 15 min to complete;
- **P#5:** The user directly navigated to the planner profile but was confused about the purpose of the “Plan” and “+ new planning” buttons. The user expected pictograms to be added to the sequence when clicked on. Once they knew how to add pictograms to the sequence, the user had no problem removing them. The user identified the calming object successfully but tried to perform more actions with the pictogram than the application allowed. The tasks took them around 14 min to complete;
- **P#6:** The user accessed the scheduler profile directly. The message “No schedules available” made them think they could not create a new schedule. The user had difficulty

modifying pictograms when planning an event and was confused by the “back” buttons. The user identified the communication notebook but didn't find any tranquilizing objects. The user suggested adding the word “steps” to the “+ new planning” button, making it “planning steps”. The tasks took them around 19 min to complete;

- **P#7:** The user navigated directly to the scheduler profile and clicked “New event” to create a new schedule. The user attempted to schedule a new event without setting a time and consultation type and struggled with selecting pictogram categories. The user was unsure why the new schedule wasn't shown on the calendar after its creation. The user interacted with the communication notebook initially and interacted with a calming object, but it caused them stress as they expected more significant interaction with it. They mentioned not identifying any reassuring objects. The tasks took them around 17 min to complete.

4.3.4 List of comments and recommendations.

The main comments and improvement recommendations are summarized as follows:

- o It is essential to improve the application in the aspects belonging to the engaging, structured, and frustration-free categories specified in the inspection with the property checklist (Appendix C). Providing timely information when acting, for example, when creating a new timetable, could help improve the application's attractiveness and engagement. Maintaining consistency in the information presented and interacting with the elements of the application would help improve the structure of the application. In addition, providing readable and structured documentation, messages, and error recovery options will help to make a frustration-free system;
- o The 43 potential problems identified by the evaluators (Appendix F) must be addressed, prioritizing the ones with greater criticality (Appendix I) and severity (Appendix J). For the main potential problems, the following recommendations are presented:
- P2 “Options use iconography and do not include text that they represent”: Information must be provided about the elements on the screen; for example, include a text saying “communications notebook” under said object/button in the user profile;
- P17 “Difficulty and inconsistency in saving an event on a specific day”: When creating a new schedule, it is not assigned to the day and time previously selected in the application. It is suggested that the selection of schedules available in the application be made through a checkbox or some similar element that helps the user visually iden-

tify that they must select it to be assigned on the desired day;

- P6 "Difficult movement of pictograms in planning": It is recommended that when selecting a pictogram in the application, it provides visual feedback to the user, such as highlighting the edges of the pictogram to indicate that it has been correctly selected for movement, like the behavior of apps in a tablet menu when long pressing and then rearranging them;
 - P14 "Delete progress on day change": It is suggested that the system warns and waits for a response from the user before changing to a different day on the calendar;
 - P16 "Visible events outside of the date": It is recommended that the application visually identifies medical appointments on dates already in the past in a distinctive way. Verifying the current date of the tablet with the date of the schedules assigned in the calendar would allow discrimination in the schedules permitted to interact;
 - P8 "Difficulty removing a pictogram": It is suggested to exchange the positions of the sections (available pictograms and green rectangle with the sequence with the added pictograms) in the creation tab of a new plan, leaving the sequence of pictograms at the bottom from the screen. When carrying out this exchange, eliminating a pictogram from the sequence is more natural since moving it up can be understood that said pictogram is returned to its origin. An alternative to deleting the pictograms can also be offered by adding a button with a "delete" icon that appears after long-pressing on the pictogram to select it.
- o The results obtained from the user test confirm the importance of solving the potential problems identified in the inspection methods. Potential problems, such as the difficulties in interacting with the pictograms, the creation, and assignment of a medical appointment in the calendar, are not intuitive, and the little feedback from the application on the elements present on the screen was confirmed by the results obtained in the controlled observation. Additionally, this test highlights the importance of providing clear messages free of interpretations, the need to differentiate between the categories of pictograms and the pictograms available to be added to the sequence, and the need to clarify the intention of the "back" buttons, for which users had difficulties in identifying them correctly.

4.3.5 UX report

After executing the methods, a comprehensive UX report document was generated. This report includes information on how the methodology was applied and presents the

complete results of this process. It is recommended that the report be organized according to the evaluation methods employed: property checklists, heuristic evaluation, and controlled observation. The report also features a section that presents an overview of the findings, underscores the potential issues identified, and proposes recommendations for UX enhancement. The text contained in the report is shown below in italics and in quotation marks to identify each of its different sections.

"The PlanTEA app achieved a satisfaction score of 68.5%, where the engaging, structured, and frustration-free stand out as categories for improvement."

"After eliminating potential duplicate issues flagged by the evaluators, the next 43 problems have been identified, as well as the heuristic that each one does not meet."

Then, Table 3 of Appendix F is included.

"Based on the severities and criticalities with the highest scores, as the next tables show," (Table 6 of Appendix I and Table 7 of Appendix J are included here) "the following recommendations are proposed for improvement:"

5 Conclusions

In the development of computer systems, it is necessary to validate the user experience as a quality factor that significantly impacts the acceptance and future adoption of the developed software. This need is even more significant when designing software for special-needs user groups. In these cases, there is a common need for specific evaluation methods and techniques adapted to the particularities of these users.

This paper describes the application of a specific methodology for UX evaluation of a software application for users with ASD called PlanTEA, which helps them plan and anticipate their visits to medical centers. This methodology comprises a three-stage validated process based on the UX/ASD model. It contributes to the field of UX evaluation methodologies since it focuses on assessing the user experience of systems used by adults with ASD by considering specific tools and techniques for effective evaluation. It utilizes a validated process with specific methods and instruments selected and adapted according to the needs of people with ASD, in contrast to the few existing that are limited to the use of generic usability analysis tools, such as adaptations of the Nielsen heuristics [9] or informal interviews with family members and caregivers [10].

Using the evaluation methodology facilitated the identification of numerous potential issues from the viewpoints of

various evaluators and ASD users. These individuals' professional and personal experiences provided diverse insights into the user experience of an individual with ASD interacting with the PlanTEA application. The support from professionals experienced in UX evaluations and experts with practical experience with individuals with ASD enabled the generation of a comprehensive report on the application's key aspects that needed improvement.

The application of the methodology in assessing the UX in the PlanTEA app proved beneficial in uncovering a range of potential problems. This study's outcomes affirm that the methods employed and proposals in the methodology complement each other, offering a broader diversity of perspectives on the potential problems identified.

As a result of the evaluation, some aspects related to the roles and access profiles of the application, as well as the navigation, help, and interaction mechanisms, such as the direct manipulation supported in the current version of the application, will be revised and modified. In addition, some functional issues that will improve the robustness of the next version of this software will be checked. Finally, its portability to other devices and access platforms and its generalization to different usage scenarios (e.g., planning and anticipating school attendance, use of public transport, shopping, etc.) will be addressed.

Providing solutions to the main potential UX problems in a future version of PlanTEA will contribute to ensuring a better user experience for people with ASD. It is believed that the improvements derived from the conducted evaluation will not only enhance the usability and accessibility of PlanTEA but will also enable people with ASD to benefit from its use in a broader range of everyday situations, thus improving their autonomy and quality of life.

Finally, the lessons learned from this experience can be summarized in the following practical guideline: the use of generic usability evaluation instruments such as SUS or TAM that are not adapted to users with ASD does not provide specific results for evaluating this methodology, and it is necessary to use other instruments adapted to the type of user of our PlanTEA application. Although the app was developed using a UCD approach, it is essential to include a UX/ASD expert at the first stages of the development to avoid such potential usability problems in the final app, and it is recommended to evaluate all the prototypes during the development as soon as possible to improve the UX of the final app. The participation in these evaluations of potential users with ASD, with characteristics that differentiate them from the rest of the population, is especially important in order to evaluate the satisfaction and perception of their experience with the evaluated application.

6 Appendix A: User test design

Below is the document's content provided to participants (ASD users) during the controlled observation. It begins with a description of PlanTEA and the scenario in which they will have to perform the tasks list indicated next.

PlanTEA is a mobile application that allows planning and anticipating the attendance of children or adults with *Autism Spectrum Disorder* (ASD) at medical consultations and facilitates communication with specialists through a communication notebook.

When using the PlanTEA app, consider the following context:

*On June 9 at 4:00 p.m., you have an appointment to get a flu shot at your health center, which **IS NOT SCHEDULED** in the PlanTEA application.*

To plan your appointment at the doctor's office, consider that you should at least: leave home, go to the doctor's office and vaccination room, wait your turn, contemplate an activity while you wait, and have the nurse tell you that it is your turn, disinfect, vaccinate, and put a patch, wait after vaccination, and finally return home.

Therefore, you must access the PlanTEA application through the **scheduler** profile using the tablet you were issued. Please perform each task and comment aloud throughout the experiment if you are comfortable.

ENTER THE PLANTEA APP (all tasks can be performed without exiting the application). Complete the following tasks:

1. First Task

Considering the context mentioned above, you must plan your appointment to get a flu shot in the PlanTEA application through the **SCHEDULER** profile. Consider that the password is **123abcd**.

- Have you found the day you should schedule your appointment? _____
- What type of consultation is the one you have indicated in the application? _____
- Have you correctly planned your appointment in the application for the day and time indicated? _____

2. Second Task

Enter the **USER** profile. The app has a calming object in case you feel overwhelmed. Answer:

- Can you find the calming object? _____
- What is this calming object? _____

7 Appendix B: Preliminary and perception questionnaires

7.1 B.1. Inspections Preliminary Questionnaire

1. Profession/Area of research
2. Years of experience in the profession/area
3. Before being a collaborator in our research, were you familiar with UX and/or usability concepts? (Yes / No)
4. If yes, how long have you been familiar with UX and/or Usability concepts?
5. Before being a collaborator in our research, were you aware of the TEA condition (Yes / No)?
6. If yes, how long have you been aware of the ASD condition?
7. Have you used applications designed for people with ASD (Yes /No)?
8. If yes, please indicate which system(s) or application(s) you have used.
9. If you answered yes to question 7, how has your experience been with these systems or applications (Very Bad / Bad / Fair / Good / Very Good)?
10. Do you use applications to plan your daily life? (Never / Almost never / Sometimes / Almost always / Always)

7.2 B.2. Inspections Perception Questionnaire

7.2.1 Answers values of a 5-level Likert scale

- 1: Very little oriented/Very difficult/Very ineffective/Totally unsatisfactory/Strongly disagree.
- 2: Poorly oriented/Difficult/Not very effective/Unsatisfactory/Disagree
- 3: Neutral
- 4: Oriented/Easy/Effective/Satisfactory/Agree
- 5: Very oriented/Very easy/Very effective/Very satisfactory/Strongly agree

7.2.2 Questions

1. Does the application do a good job of orienting the user during the interaction?
2. How is the interaction with the application?
3. How effectively have your interactions with the application achieved the stated objectives?
4. What is your degree of satisfaction after interacting with the application?
5. Would you recommend using the PlanTEA application to plan medical appointments and communicate with specialists?

6. Do you consider that the application is attractive for users with ASD?
7. Do you consider that the application generates an environment of trust for users with ASD?
8. Do you consider that the application elements are presented orderly, structured, coherent, and consistent throughout the interaction?
9. Do you consider that the interactions with the application consider the needs of users with ASD?
10. Do you consider that the elements of the application (text, icons, processes, sounds...), as well as the interactions with them, are familiar and like those that can be found in real life?
11. Do you consider that the application is customizable?
12. Do you consider that the application is designed considering the sensory characteristics of users with ASD (sight, hearing, and touch)?
13. Do you consider that the application retains the attention of users with ASD?
14. Do you consider that the application adequately manages time?
15. Do you consider the application avoiding users' frustrations with ASD during interaction?
16. What positive aspects of the PlanTEA application would you highlight?
17. What negative aspects of the PlanTEA application would you highlight?

7.3 B.3. Controlled Observation Preliminary Questionnaire.

1. Gender:

- Female ()
Male ()

2. Age:

3. Completed education level:

- No studies: ()
Compulsory Secondary Education (ESO): ()
High School: ()
Vocational Training: ()
University: ()
Other:

4. Do you plan your medical appointments in advance? (Never/Almost never/Sometimes/Almost always/Always)
5. Do you find communicating with the doctor during a medical visit challenging? (Never/Almost never/Sometimes/Almost always/Always)
6. What is your level of satisfaction when you return from the doctor's office? (Completely dissatisfied/Dissatisfied/Somewhat satisfied/Satisfied/Completely satisfied)

7. How often do you use tablets, cell phones, and/or computers daily?

Never—1 day a week—More than two days a week—
More than four days a week—All days of the week

8. How often do you use applications designed for people with ASD? (Never/Almost never/Sometimes/Almost always/Always).

7.4 B.4. Controlled Observation Perception Questionnaire

1. Was it challenging to complete the tasks indicated? (Very difficult/Difficult/Normal/Easy/Very easy)
2. Was it challenging to find the required information? (Very difficult/Difficult/Normal/Easy/Very easy)
3. Did the application provide you with enough information to perform the tasks? (Never/Rarely//Half the time/Most of the time/Always)
4. Is the information provided by the application understandable? (Definitely not/Probably not/Neutral/Probably yes/Definitely yes)
5. Has the application helped/guided you in knowing what to do when planning your medical consultations? (Strongly disagree/Disagree/Neutral/Agree/Strongly Agree)
6. Did the application help/guide you on what to do during your dental visit? (Strongly disagree/Disagree/Neutral/Agree/Strongly Agree)
7. Did you find it easy to use the application? (Very complicated/Complicated/Normal/Easy/Very easy)
8. Did you feel overwhelmed by the information provided by the application? (Definitely not/Probably not/Neutral/Probably yes/Definitely yes)
9. Are the images, icons, and symbols used in the application easy to understand? (Definitely not/Probably not/Neutral/Probably yes/Definitely yes)
10. Has the application allowed you to customize any aspect? (Definitely not/Probably not/Neutral/Probably yes/Definitely yes)
11. Have you been bothered by any application's visual and/or sound elements? (Strongly disagree/Disagree/Neutral/Agree/Strongly Agree)
12. Have you been distracted during the interaction with the application? (Strongly disagree/Disagree/Neutral/Agree/Strongly Agree)
13. When you made a mistake, did the application help you to fix it? (Strongly disagree/Disagree/Neutral / Agree/Strongly Agree)
14. Would you use PlanTEA again to plan your following medical appointments? (Strongly disagree/Disagree/Neutral/Agree/Strongly Agree)

15. Would you use PlanTEA again to attend your following medical appointments? (Strongly disagree/Disagree/Neutral/Agree/Strongly Agree)
16. What did you like most about PlanTEA?
17. What did you like least about PlanTEA?

8 Appendix C. Property Checklist

List of items that compose the property checklist [33]:

1. *Engaging*: The system commits the user to interact with it.
 - 1.1. The system delivers constant feedback to the user clearly and concisely. This must be through text, visual elements, or audio.
 - 1.2. The system rewards the user for positive actions or good performance.
 - 1.3. The system has a history of the actions performed by the user.
2. *Predictable*: The system has a predictable environment, generating user trust.
 - 2.1. The system does not have sudden and unexpected actions.
 - 2.2. The content of the system is predictable and consistent.
 - 2.3. The system allows repeating actions, tasks, or activities.
3. *Structured*: The system is structured.
 - 3.1. The system's navigational setup, aesthetics, and content are structured and consistent.
 - 3.2. Navigation in the system is simple and logical.
4. *Interactive*: The system generates interactions considering the users' characteristics, affinities, and needs, as well as their difficulties in social interactions.
 - 4.1. Each task has a clear and explicit objective.
 - 4.2. Instructions on the system are clear, simple, brief, and context-appropriate.

4.3. The system provides elements to minimize the user's memory load (for example, grouping and delivering the necessary and concise information to the user).

5. *Generalizable*: The system is familiar enough to facilitate generalizing skills like real life.

5.1. The system's Activities, tasks, or information are based on previously learned activities, tasks, or information.

5.2. The interaction with the system is familiar and like the users' real lives.

6. *Customizable*: The system can be customized considering the needs, abilities, and preferences of people with ASD.

6.1. The system allows users and tutors to customize aspects quickly, easily, and effortlessly. (e.g., turn off sounds, configure the level and intensity of the sounds, modify the colour palette, font type, size, layout, and activity times).

6.2. The system has a predefined basic configuration that considers the characteristics, affinities, and needs of users with ASD.

7. *Sense-aware*: The system considers the senses of users with ASD.

7.1. The system provides information to the user through multimedia texts, among others.

7.2. The visual and sound elements are clear, meaningful, functional, non-disruptive, and legible.

7.3. The actions and states of the system are clear and straightforward.

7.4. The system interface is clear, simple, and minimalist.

7.5. Texts have clear messages, readable font size, and type, and are free of rhetorical figures.

8. *Attention Retaining*: The system retains users' attention and manages time appropriately.

8.1. The system responds to user actions quickly and without delays.

8.2. The system provides dynamic stimuli such as animations and/or controlled music to attract users' attention.

8.3. The system views do not provide distracting elements.

9. *Frustration-free*: The system tries to avoid the frustration of its users during their interaction.

9.1. The system allows you to confirm, cancel, or repair unwanted actions.

9.2. The system displays error messages in plain language, accurately indicates the problem, and constructively suggests how to avoid such errors.

9.3. The documentation and help provided by the system are provided in a visual, textual, concrete, not extensive, and structured way.

9 Appendix D: Heuristics for UX evaluation in people with ASD

Table 1 shows the heuristics list used in the PlanTEA inspection, specified by an identifier (ID), a name, and a definition.

10 Appendix E: Evaluation sheet for researchers

Table 2 briefly specifies the inputs, execution steps, outputs, and expected time to assess the controlled observation. There is also space to include any relevant remarks or comments.

11 Appendix F: UX problems detected in PlanTEA

Table 3 shows the list of potential PlanTEA problems found by each evaluator, including a definition, some comments and explanations about it, the location (mode or display) where it was found, and the failing heuristic.

12 Appendix G: Unfulfilled heuristics of PlanTEA

Table 4 summarizes the identified potential problems associated with unfulfilled heuristics, as well as the number of problems related to each one.

13 Appendix H: Severity, frequency, and criticality of PlanTEA problems assigned by each evaluator

Table 5 shows the results obtained after each evaluator (Ev) assigned each potential problem a severity (S), frequency (F), and criticality (C) value on a 5-level scale. The last columns show each aspect's mean and standard deviation for each problem.

14 Appendix I: Potential problems ordered by criticality

Table 6 shows the potential problems found, arranged in descending order based on the critical averages obtained.

15 Appendix J: Potential problems ordered by severity

Table 7 shows the potential problems found, arranged in descending order based on the severity averages obtained.

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Data availability Datasets generated and analyzed are available from the corresponding author on reasonable request.

Declarations

Conflict of interest The authors declare no competing interests.

Ethical approval The Social Research Ethics Committee (SREC) of the University of Castilla-La Mancha, Spain, evaluated and verified compliance with the ethical standards for social research in the study described (CEIS-648868-S1J7).

Consent for publication All authors have consented to publication.

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